

# Verified Integer Rational Database

## DATA-4: Includes Cabibbo Doublet Extension

**Registry:** [\[HOWL-DATA-4-2026\]](#)

**Series Path:** [\[HOWL-DATA-1-2026\]](#) → [\[HOWL-DATA-2-2026\]](#) → [\[HOWL-DATA-3-2026\]](#) → [\[HOWL-DATA-4-2026\]](#)

**DOI:** 10.5281/zenodo.19674257

**Date:** April 1 2026

**Domain:** Cross Domain Data

**Status:** Documentation

**AI Usage Disclosure:** Only the top metadata, figures, refs and final copyright sections were edited by the author. All paper content was LLM-generated using Anthropic's Claude Opus 4.6.

**Backed by:** data\_4.py (38/38 checks), DATA-3 (32/32 checks inherited)

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## Abstract

DATA-4 is the verified successor to DATA-3. It inherits all 123 entries and all 32 consistency checks with zero numerical changes. It adds 23 new entries in two categories: 6 STAGED entries for the Cabibbo Doublet parameters (entries 124-129, Type G — bounded but not yet measured) and 17 GUT/unification parameters (Section N, Type D — derived from Level 1 exact Fraction arithmetic). It formalizes the lattice ratio independence annotation as Finding 15: the FLAG lattice QCD mass ratios ( $m_c/m_s$ ,  $m_b/m_c$ ,  $m_u/m_d$ ) are independent measurements evaluated at a common renormalization scale, not derivable from the individual PDG quark masses evaluated at different scales, with discrepancies up to 28% from the scale mismatch. It adds 6 new GUT verification checks (Group G, all exact), bringing the total to 38/38 PASS. It adds a computation traceability map linking DATA-4 entries to every PHYS and MATH paper in the series. DATA-4 contains 146 entries total: 123 inherited from DATA-3, 6 staged Cabibbo Doublet, and 17 GUT parameters. DATA-4 is the sole data reference for all future HOWL computation. DATA-3 is retired.

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## 1. Relationship to DATA-3

DATA-3 collected 123 entries (107 original Q335 conversions from DATA-2 plus 16 Koide-derived quantities) and verified them with 32 internal consistency checks, all passing. DATA-3 also noted — in script output but not as a formal finding — that three lattice QCD mass ratios are independent of the individual PDG quark masses.

DATA-4 inherits every entry and every check from DATA-3 with zero numerical changes. The 32 inherited checks produce identical results (the data didn't change). What DATA-4 adds:

Finding 15 formalized: the lattice ratio independence annotation, promoted from a script note to a registered finding with a registry tag.

Section L: 6 STAGED entries for the Cabibbo Doublet parameters ( $M_{VL}$ ,  $\theta_{14}$ ,  $\theta_{24}$ ,  $\theta_{34}$ ,  $\delta_1$ ,  $\delta_2$ ), with constraint windows from PHYS-16 and PHYS-19. These are the first Type G entries in the series — parameters identified by the theory but not yet measured.

Section N: 17 GUT and unification parameters, all Level 1 exact Fractions. SM beta coefficients, Cabibbo Doublet beta shifts, modified betas, gap ratios (SM 218/115, Cabibbo Doublet 38/27, MSSM 7/5), the measured gap ratio derived from Level 2 couplings, the two-loop  $b_{ij}$  matrix, and two-loop unification results.

Group G: 6 new verification checks on the GUT parameters (all exact).

Group T: computation traceability map linking entries to papers.

The promotion from DATA-3 to DATA-4 means: annotation formalized, new particle staged, unification parameters recorded, traceability added.

## 2. The Lattice Ratio Independence — Finding 15

Entries D9 ( $m_c/m_s = 11.783$ ), D10 ( $m_b/m_c = 4.578$ ), and D11 ( $m_u/m_d = 0.485$ ) are independent measurements from lattice QCD (FLAG averages) evaluated at a common renormalization scale. They are NOT derivable from the individual PDG quark masses (D1-D5), which are evaluated at different scales:  $m_u$ ,  $m_d$ ,  $m_s$  at  $\mu = 2$  GeV MS-bar;  $m_c$  at  $\mu = m_c \approx 1.27$  GeV;  $m_b$  at  $\mu = m_b \approx 4.18$  GeV.

The discrepancies if naively compared by dividing PDG masses:

| Ratio     | PDG Division         | Lattice (FLAG) | Discrepancy |
|-----------|----------------------|----------------|-------------|
| $m_c/m_s$ | $1273/93.5 = 13.615$ | 11.783         | 15.5%       |
| $m_b/m_c$ | $4183/1273 = 3.286$  | 4.578          | 28.2%       |
| $m_u/m_d$ | $2.16/4.70 = 0.460$  | 0.485          | 5.2%        |

This is a renormalization scale mismatch, not a database error. QCD running changes quark mass ratios because each mass runs differently under the strong coupling. The  $m_b/m_c$  discrepancy (28%) is the largest because  $m_b$  and  $m_c$  are evaluated at the most widely separated scales. The  $m_u/m_d$  discrepancy (5.2%) is the smallest because both are evaluated near 2 GeV.

This matters operationally: a future session dividing  $m_c/m_s$  from entries D4 and D3 and comparing to entry D9 will see a 15% discrepancy. Without this annotation, they will conclude the database is corrupted. It is not.

Finding 15 registered: [?]

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### 3. Cabibbo Doublet Staged Entries — Section L

Six parameters of the Cabibbo Doublet (PHYS-15, PHYS-16, PHYS-19) are registered as STAGED entries. Type G means: the parameter is identified by theory, bounded by current experiment, but not measured with precision sufficient for a definitive database value and Q335 numerator.

**Entry 124:  $M_{VL}$  (Cabibbo Doublet mass).** Window: 1,500,000 - 6,000,000 MeV (1.5 - 6.0 TeV). Lower bound from LHC pair production exclusion. Upper bound from CKM perturbativity and anomaly fit. Sources: PHYS-16, PHYS-19.

**Entry 125:  $\sin \theta_{14} \approx |V_{ub}|$  (1st-generation mixing angle).** Estimate:  $\sim 0.045$  from CKM first-row deficit. Source: PHYS-19 (Belfatto, Berezhiani 2020).

**Entry 126:  $\theta_{24}$  (2nd-generation mixing angle).** Constrained by kaon physics ( $K^0$ - $\bar{K}^0$  mixing, NA62). No point estimate; bounded from above.

**Entry 127:  $\sin \theta_{34}$  (3rd-generation mixing angle).** From  $A_{FB}^b$  fit at LEP Z-pole. Source: PHYS-19.

**Entry 128:  $\delta_1$  (new CP phase 1).** Constrained by neutron EDM  $< 10^{-26}$  e-cm.

**Entry 129:  $\delta_2$  (new CP phase 2).** Constrained by B-meson CP asymmetries (LHCb, Belle II).

When any parameter is measured, the entry transitions from Type G to Type M with a specific value, uncertainty, and Q335 numerator. The entry number is preserved — no renumbering.

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### 4. GUT and Unification Parameters — Section N

Seventeen parameters from the Session 3 GUT and unification analysis, all stored as exact Fractions:

**N1-N3: SM one-loop beta coefficients (Level 1).**  $b_1 = 41/10$ ,  $b_2 = -19/6$ ,  $b_3 = -7$ . These are the exact rational beta function coefficients for  $SU(3) \times SU(2) \times U(1)$  with the SM particle content, computed from Dynkin indices.

**N4-N6: Cabibbo Doublet beta shifts (Level 1).**  $\Delta b_1 = 1/15$ ,  $\Delta b_2 = 1$ ,  $\Delta b_3 = 1/3$ . From the Dynkin indices of the (3,2,1/6) VL representation.

**N7-N9: Modified beta coefficients (Level 1).**  $b_{1\_mod} = 25/6$ ,  $b_{2\_mod} = -13/6$ ,  $b_{3\_mod} = -20/3$ . The SM betas plus the Cabibbo Doublet shifts.

**N10-N12: Gap ratios (Level 1).** SM gap ratio =  $218/115 = 1.896$  (N10). Cabibbo Doublet gap ratio =  $38/27 = 1.407$  (N11). MSSM gap ratio =  $7/5 = 1.400$  (N12). All exact Fractions.

**N13: Measured gap ratio (Derived from Level 2).** Computed from  $\alpha^{-1}$ ,  $\sin^2\theta_W$ ,  $\alpha_s$  through GUT normalization:  $(1/\alpha_1 - 1/\alpha_2)/(1/\alpha_2 - 1/\alpha_3) \approx 1.358$ . This is the confrontation value — the target the SM misses by 40% and the Cabibbo Doublet approaches to 3.6%.

**N14: Two-loop SM  $b_{ij}$  matrix (Level 1).** The  $3 \times 3$  matrix of two-loop beta function coefficients from Machacek-Vaughn (1983) and Luo-Xiao (hep-ph/0207271), all exact Fractions:

|       | U(1)   | SU(2) | SU(3) |
|-------|--------|-------|-------|
| U(1)  | 199/50 | 27/10 | 44/5  |
| SU(2) | 9/10   | 35/6  | 12    |
| SU(3) | 11/10  | 9/2   | -26   |

**N15-N17: Two-loop unification results (Derived).** At  $M_{VL} = 500$  GeV (closest approach): one-loop miss  $\Delta = -1.17$  (N15), two-loop miss  $\Delta = -0.40$  (N16), two-loop improvement = 66% (N17). The two-loop correction reduces the unification miss by two-thirds.

## 5. Verification

### 5.1 Inherited Checks (Groups A-E): 32/32 PASS

All 32 DATA-3 checks are re-run on the unchanged data and produce identical results. The full results from the data\_4.py script:

| Group                   | Tests | Pass | Content                                                           |
|-------------------------|-------|------|-------------------------------------------------------------------|
| A: Mass ratios          | 9     | 9    | Stored ratios vs computed from individual masses                  |
| B: Analytical constants | 8     | 8    | Q335 identities and mpmath cross-checks at 101+ digits            |
| C: Physical relations   | 4     | 4    | $R_\infty$ , $a_0$ , $\mu_0$ from formulas; m D from constituents |
| D: Koide derived        | 5     | 5    | K1-K16 recomputed from source masses                              |

| Group                 | Tests | Pass | Content                                                            |
|-----------------------|-------|------|--------------------------------------------------------------------|
| E: SI exact constants | 6     | 6    | c, h, e, k B, N A, $\Delta$<br>$\nu$ Cs at exact SI<br>2019 values |

No changes from DATA-3. Every test passes at the same digit count.

## 5.2 New Checks (Group G): 6/6 PASS

Six new GUT verification checks, all exact:

| Test | Relation                | Result |
|------|-------------------------|--------|
| G1   | SM gap ratio = 218/115  | EXACT  |
| G2   | SM+VL gap ratio = 38/27 | EXACT  |
| G3   | MSSM gap ratio = 7/5    | EXACT  |
| G4   | Modified $b_1 = 25/6$   | EXACT  |
| G5   | Modified $b_2 = -13/6$  | EXACT  |
| G6   | Modified $b_3 = -20/3$  | EXACT  |

All six are identities between exact Fractions. The gap ratios are computed from the beta coefficients by division:  $(b_1-b_2)/(b_2-b_3)$ . The modified betas are computed by addition:  $b_{\text{SM}} + \Delta b_{\text{VL}}$ . Every step is Fraction arithmetic with zero floating-point contamination.

## 5.3 Combined Result

**38 tests. 38 PASS. 0 FAIL.**

Inherited from DATA-3: 32 checks (Groups A-E). New in DATA-4: 6 checks (Group G). The lattice annotation (Group F) and Cabibbo Doublet staging (Group H) are structural additions, not numerical checks — there are no values to verify for staged entries.

## 6. Database Content

DATA-4 contains 146 entries across the following sections:

| Section            | Type         | Count | Content                                                                                                             |
|--------------------|--------------|-------|---------------------------------------------------------------------------------------------------------------------|
| A: SI fundamental  | E (exact)    | 7     | c, h, e, k B, N A, $\Delta$<br>$\nu$ Cs, K cd                                                                       |
| B: CODATA measured | M (measured) | 13    | $\alpha^{-1}$ , m e, m $\mu$ , m $\tau$ ,<br>m p, ratios, a e, a $\mu$<br>, $\sin^2\theta_W$ , $\alpha_s$ , $\mu_0$ |

| Section                    | Type           | Count      | Content                                                                                                                            |
|----------------------------|----------------|------------|------------------------------------------------------------------------------------------------------------------------------------|
| C: Electroweak             | M              | 6          | M Z, $\Gamma$ Z, M W, m t, m H, G F                                                                                                |
| D: Quarks + CKM            | M              | 11         | m u-m b, sin $\theta_{12}$ -sin $\theta_{13}$ , lattice ratios (D9-D11 INDEPENDENT)                                                |
| E: Nuclear/hadron          | M              | 8          | m n, m $\pi$ , m K, m D, m He4, E D                                                                                                |
| F: Spectroscopy            | M              | 1          | H 1S-2S frequency                                                                                                                  |
| G: Q335 analytical         | A (analytical) | 14         | $\pi$ , e, $\ln 2$ , $\sqrt{2}$ - $\sqrt{7}$ , $\varphi$ , $\zeta(3)$ , $\zeta(5)$ , $\pi^2$ , $\zeta(2)$ , $R_2$ , $R_4$ , $2\pi$ |
| K: Mass ratios + Koide     | M/K            | 8          | m $\mu$ /m e, m $\tau$ /m e, etc., Koide K                                                                                         |
| L: Cabibbo Doublet         | G (staged)     | 6          | M VL, $\theta_{14}$ , $\theta_{24}$ , $\theta_{34}$ , $\delta_1$ , $\delta_2$                                                      |
| N: GUT parameters          | D (derived)    | 17         | Betas, shifts, gap ratios, b ij, two-loop results                                                                                  |
| (remaining DATA-3 entries) | Various        | 56         | Engineering specs, clock freqs, Koide addendum                                                                                     |
| <b>Total</b>               |                | <b>146</b> |                                                                                                                                    |

Type codes: E = exact by definition, M = measured, A = analytical (computed to arbitrary precision), S = standard nominal, K = Koide derived, G = staged (bounded, not measured), D = derived (Level 1 arithmetic on Level 2 inputs or pure Level 1).

## 7. Computation Traceability Map

Which DATA-4 entries feed which computations, from the data\_4.py traceability section:

| Paper   | DATA-4 Entries Used                                | Computation                                          |
|---------|----------------------------------------------------|------------------------------------------------------|
| PHYS-9  | B1( $\alpha^{-1}$ ), G1( $\pi$ ), G8( $\zeta(3)$ ) | $\alpha \rightarrow a e$ via QED series              |
| PHYS-12 | B1, B11, C6, C4, C5, C1, B12                       | 7 EW inputs $\rightarrow$ 11 observables             |
| PHYS-13 | B1, B11, B12                                       | 3 couplings $\rightarrow$ gap ratio 218/115 vs 1.358 |
| PHYS-15 | B1, B11, B12                                       | Elimination cascade $\rightarrow$ (3,2,1/6)          |
| PHYS-17 | N1-N3 (SM betas)                                   | Generation democracy, boson problem                  |

| Paper   | DATA-4 Entries Used                       | Computation                                             |
|---------|-------------------------------------------|---------------------------------------------------------|
| PHYS-18 | N4-N6 (VL shifts)                         | $Y=1/6$ asymmetry, $1/Y^2$ scaling                      |
| PHYS-19 | D6-D8 (CKM), B11                          | Three anomalies $\rightarrow$ (3,2,1/6)                 |
| PHYS-20 | N7-N11 (modified betas, gap ratio)        | M GUT $\rightarrow$ proton decay $\tau \sim 10^{34-35}$ |
| PHYS-22 | G10( $\pi^2$ ), G8( $\zeta(3)$ ), G3(ln2) | $A_2 = 197/144 + (3/4)\zeta(3) + R_4 c$ geom            |
| PHYS-23 | B2-B4 (lepton masses), D1-D5 (quarks)     | Koide K all three sectors                               |
| MATH-6  | G1-G14 (Q335 basis)                       | PSLQ 82/82 null, Bessel zeros                           |

When a future PDG update changes any entry, this map identifies every computation that must be re-verified. If  $\alpha_s$  changes: PHYS-12, 13, 15 are affected. If  $m_\tau$  improves: PHYS-23 Koide checks sharpen. If  $\sin^2\theta_W$  shifts: PHYS-12, 13, 15, 19 are affected and the measured gap ratio (N13) recomputes.

## 8. What DATA-4 Certifies

**Everything DATA-3 certified, unchanged.** Internal consistency (32/32), Q335 fidelity (101+ digits), physical consistency (CODATA web at 11+ digits), independence structure (107 original sources).

**Plus GUT parameter consistency.** The 6 new Group G checks verify that gap ratios and modified betas are exact Fraction consequences of the stored SM betas and Cabibbo Doublet shifts. 6/6 exact.

**Plus structural completeness.** The Cabibbo Doublet parameters are formally registered with entry numbers, constraint windows, and transition protocol. The GUT parameters that drive the unification analysis are stored as exact Fractions with complete provenance. The computation traceability map links data to papers.

## 9. What DATA-4 Does Not Change

No inherited entry value changes. No Q335 numerator changes. No uncertainty changes. No type reclassification of existing entries. The 32 inherited check results are verbatim from DATA-3. DATA-4 is additive: it adds entries, checks, and structure. It does not correct, modify, or supersede any numerical content from DATA-3.

## 10. Operational Rule

DATA-4 is the sole data reference for all future HOWL computation. DATA-3 is retired. Any script or paper that previously referenced DATA-3 now references DATA-4. The numerical content of entries 1-123 is identical; the distinction is the additional entries, checks, and structure.

When new measurements become available — improved  $m_\tau$  from Belle II, LHC VL quark mass, updated PDG world averages, new CODATA adjustment — they enter as DATA-5 with the same verification protocol: every affected cross-check must pass before the new version is accepted. No DATA version is ever edited after publication.

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## Appendix: Verification Script Output

From data\_4.py, 38/38 checks pass:

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HOWL-DATA-4: VERIFIED DATABASE WITH CABIBBO DOUBLET EXTENSION
=====
```

GROUP A: MASS RATIO IDENTITIES

```
[PASS] A1: m_p/m_e      12.2 digits (need 11)
[PASS] A2: m_mu/m_e     21.0 digits (need 10)
[PASS] A3: m_tau/m_e     6.3 digits (need 6)
[PASS] A4: m_tau/m_mu    5.8 digits (need 5)
[PASS] A5: m_n/m_p      20.0 digits (need 11)
[PASS] A6: M_W/M_Z       7.2 digits (need 6)
[PASS] A7: m_n - m_p     exact      (need 8)
[PASS] A8: m_H/M_Z       5.6 digits (need 5)
[PASS] A9: m_t/M_Z       5.9 digits (need 5)
```

GROUP B: ANALYTICAL CONSTANT IDENTITIES (Q335 basis)

```
[PASS] B1: R2 =  $\pi/4$       102.0 digits (need 100)
```



[PASS] B2:  $R_4 = \pi^2/32$  102.4 digits (need 100)

[PASS] B3:  $2\pi = 8R_2$  exact (need 100)

[PASS] B4:  $\zeta(2) = \pi^2/6$  101.5 digits (need 99)

[PASS] B5:  $\alpha/\pi$  identity 13.9 digits (need 12)

[PASS] B6:  $\varphi = (1+\sqrt{5})/2$  101.4 digits (need 100)

[PASS] B7:  $\pi^2 = \pi \times \pi$  101.7 digits (need 99)

[PASS] B8:  $2\pi = 2 \times \pi$  exact (need 100)

GROUP C: PHYSICAL RELATIONS (SI 2019)

[PASS] C1:  $R_\infty = \alpha^2 m_e c / (2h)$  11.3 digits (need 11)

[PASS] C2:  $a_0 = \hbar / (m_e c \alpha)$  11.4 digits (need 11)

[PASS] C3:  $\mu_0 = 2ah / (ce^2)$  11.9 digits (need 11)

[PASS] C4:  $m_D = m_p + m_n - E_D$  9.9 digits (need 8)

[INFO] C5: H 1S-2S soft check 3.3 digits (expected ~5)

GROUP D: KOIDE DERIVED ENTRIES

[PASS] D1:  $K(e, \mu, \tau)$  10.3 digits (need 6)

[PASS] D2:  $a(\text{leptons})$  10.2 digits (need 6)

[PASS] D3:  $M(\text{leptons})$  4.6 digits (need 4)

[PASS] D4:  $K(u, c, t)$  10.5 digits (need 3)

[PASS] D5:  $K(d, s, b)$  10.4 digits (need 3)

GROUP E: EXACT SI CONSTANTS

[PASS] E1:  $h \times \Delta \nu_{\text{Cs}}$  exact

[PASS] E2:  $c$  exact

[PASS] E3:  $h$  exact

[PASS] E4: e exact

[PASS] E5: k\_B exact

[PASS] E6: N\_A exact

GROUP F: LATTICE RATIO INDEPENDENCE – FINDING 15

m\_c/m\_s: PDG 13.615, lattice 11.783 (15.5%)

m\_b/m\_c: PDG 3.286, lattice 4.578 (28.2%)

m\_u/m\_d: PDG 0.460, lattice 0.485 (5.2%)

Renormalization scale mismatch, NOT database error.

GROUP G: GUT AND UNIFICATION VERIFICATION (NEW)

[PASS] G1: SM gap ratio = 218/115 exact

[PASS] G2: SM+VL gap ratio = 38/27 exact

[PASS] G3: MSSM gap ratio = 7/5 exact

[PASS] G4: Modified  $b_1$  = 25/6 exact

[PASS] G5: Modified  $b_2$  = -13/6 exact

[PASS] G6: Modified  $b_3$  = -20/3 exact

GROUP H: CABIBBO DOUBLET STAGED ENTRIES

Entry 124: M\_VL STAGED 1.5 – 6.0 TeV

Entry 125:  $\sin(\theta_{14})$  STAGED ~0.045 (CKM deficit)

Entry 126:  $\theta_{24}$  STAGED (kaon physics)

Entry 127:  $\sin(\theta_{34})$  STAGED ( $A_{FB}^b$  fit)

Entry 128:  $\delta_1$  STAGED (nEDM constrained)

Entry 129:  $\delta_2$  STAGED (B physics constrained)

GROUP T: COMPUTATION TRACEABILITY MAP

|                                                   |                              |                                           |
|---------------------------------------------------|------------------------------|-------------------------------------------|
| PHYS-9                                            | B1, G1, G8                   | alpha -                                   |
| > a_e via QED series                              |                              |                                           |
| PHYS-12                                           | B1, B11, C6, C4, C5, C1, B12 | 7 EW inputs -                             |
| > 11 observables                                  |                              |                                           |
| PHYS-13                                           | B1, B11, B12                 | 3 couplings -                             |
| > gap ratio                                       |                              |                                           |
| PHYS-15                                           | B1, B11, B12                 | Elimination cascade -                     |
| > (3,2,1/6)                                       |                              |                                           |
| PHYS-17                                           | N1-N3                        | Generation democracy, boson problem       |
| PHYS-18                                           | N4-N6                        | Y=1/6 asymmetry, 1/Y <sup>2</sup> scaling |
| PHYS-19                                           | D6-D8, B11                   | Three anomalies -                         |
| > (3,2,1/6)                                       |                              |                                           |
| PHYS-20                                           | N7-N11                       | M_GUT -> proton decay                     |
| PHYS-22                                           | G10, G8, G3                  | A <sub>2</sub> decomposition              |
| PHYS-23                                           | B2-B4, D1-D5                 | Koide K all three sectors                 |
| MATH-6                                            | G1-G14                       | PSLQ 82/82 null                           |
| Total: 38 tests, 38 PASS, 0 FAIL                  |                              |                                           |
| Sections A-K (inherited from DATA-3): 123 entries |                              |                                           |
| Section L (Cabibbo Doublet, STAGED): 6 entries    |                              |                                           |
| Section N (GUT parameters, Level 1): 17 entries   |                              |                                           |
| Total: 146 entries                                |                              |                                           |
| +=====+                                           |                              |                                           |
|                                                   | ALL TESTS PASS               |                                           |
|                                                   |                              |                                           |
|                                                   | DATA-4 DECLARATION:          |                                           |

```

| DATA-3 (123 entries, 32/32 checks) inherited. |
| Finding 15 (lattice independence) formalized. |
| 6 Cabibbo Doublet parameters STAGED (entries 124- |
129). |
| 17 GUT/unification parameters added (Section N). |
| 6 new GUT verification checks added (Group G). |
| Computation traceability map added (Group T). |
|
| DATA-3 is retired. |
| All future HOWL computation references DATA- |
4. |
+=====+

```

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*DATA-4: 146 entries, 38 cross-checks, 38 pass. The Cabibbo Doublet is staged. The integers are verified. The foundation extends. Published April 1, 2026. This paper is never edited after publication.*

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=====
HOWL-DATA-4: VERIFIED DATABASE WITH CABIBBO DOUBLET EXTENSION
=====

```

```

GROUP A: MASS RATIO IDENTITIES
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```

```

[PASS] A1: m_p/m_e direct vs m_p / m_e

```

```

Computed: 1.836152673431236e+03

```

```

Stored: 1.836152673430000e+03

```

Relative diff: 6.73e-13 (12.2 digits, need 11)

[PASS] A2:  $m_\mu/m_e$  direct vs  $m_\mu / m_e$

Computed: 2.067682827084672e+02

Stored: 2.067682827084672e+02

Relative diff: 1.01e-21 (21.0 digits, need 10)

[PASS] A3:  $m_\tau/m_e$  direct vs  $m_\tau / m_e$

Computed: 3.477228275323682e+03

Stored: 3.477230000000000e+03

Relative diff: 4.96e-07 (6.3 digits, need 6)

[PASS] A4:  $m_\tau/m_\mu$  direct vs  $m_\tau / m_\mu$

Computed: 1.681702933242618e+01

Stored: 1.681700000000000e+01

Relative diff: 1.74e-06 (5.8 digits, need 5)

[PASS] A5:  $m_n/m_p$  direct vs  $m_n / m_p$

Computed: 1.001378419463362e+00

Stored: 1.001378419463362e+00

Relative diff: 9.40e-21 (20.0 digits, need 11)

[PASS] A6:  $M_W/M_Z$  direct vs  $M_W / M_Z$

Computed: 8.813610622496918e-01

Stored: 8.813610000000000e-01

Relative diff: 7.06e-08 (7.2 digits, need 6)

[PASS] A7:  $m_n - m_p$  direct vs stored difference

Computed: 1.293332510000000e+00

```

Stored: 1.293332510000000e+00
Agreement: EXACT
[PASS] A8: m_H/M_Z direct vs stored ratio
Computed: 1.372993696511368e+00
Stored: 1.372990000000000e+00
Relative diff: 2.69e-06 (5.6 digits, need 5)
[PASS] A9: m_t/M_Z direct vs stored ratio
Computed: 1.892472222100373e+00
Stored: 1.892470000000000e+00
Relative diff: 1.17e-06 (5.9 digits, need 5)
GROUP B: ANALYTICAL CONSTANT IDENTITIES (Q335 basis)
-----
[PASS] B1: R2 = pi/4 (Q335 vs mpmath@120)
Q335: 7.853981633974483e-01
mpmath: 7.853981633974483e-01
Relative diff: 9.13e-103 (102.0 digits, need 100)
[PASS] B2: R4 = pi^2/32 (Q335 vs mpmath@120)
Q335: 3.084251375340424e-01
mpmath: 3.084251375340424e-01
Relative diff: 3.79e-103 (102.4 digits, need 100)
[PASS] B3: 2pi = 8*R2 (exact within Q335)
Computed: 6.283185307179586e+00
Stored: 6.283185307179586e+00

```

Agreement: EXACT

[PASS] B4:  $\zeta(2) = \pi^2/6$  (Q335 numerators)

Computed: 1.644934066848226e+00

Stored: 1.644934066848226e+00

Relative diff: 2.90e-102 (101.5 digits, need 99)

[PASS] B5:  $\alpha/\pi$  stored vs  $1/(\alpha_{\text{inv}} * \pi)$

Computed: 2.322819464195329e-03

Stored: 2.322819464195300e-03

Relative diff: 1.24e-14 (13.9 digits, need 12)

[PASS] B6:  $\phi = (1 + \sqrt{5})/2$  (Q335)

Computed: 1.618033988749895e+00

Stored: 1.618033988749895e+00

Relative diff: 4.42e-102 (101.4 digits, need 100)

[PASS] B7:  $\pi^2$  stored vs  $\pi * \pi$  (Q335)

Computed: 9.869604401089358e+00

Stored: 9.869604401089358e+00

Relative diff: 2.21e-102 (101.7 digits, need 99)

[PASS] B8:  $2\pi$  stored vs  $2 * \pi$  (exact)

Computed: 6.283185307179586e+00

Stored: 6.283185307179586e+00

Agreement: EXACT

GROUP C: PHYSICAL RELATIONS (SI 2019)

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[PASS] C1:  $R_{\text{inf}} = \alpha^2 m_e c / (2h)$ 
From  $\alpha, m_e, c, h$ : 1.097373156809536e+07
Stored  $R_{\text{inf}}$ : 1.097373156815700e+07
Relative diff: 5.62e-12 (11.3 digits, need 11)

[PASS] C2:  $a_0 = \hbar / (m_e c \alpha)$ 
From  $\hbar, m_e, c, \alpha$ : 5.291772105462769e-11
Stored  $a_0$ : 5.291772105440000e-11
Relative diff: 4.30e-12 (11.4 digits, need 11)

[PASS] C3:  $\mu_0 = 2 * \alpha * \hbar / (c * e^2)$ 
From  $\alpha, \hbar, c, e$ : 1.256637061268226e-06
Stored  $\mu_0$ : 1.256637061270000e-06
Relative diff: 1.41e-12 (11.9 digits, need 11)

[PASS] C4:  $m_D = m_p + m_n - E_D$ 
From  $m_p + m_n - E_D$ : 1.875612945230000e+03
Stored  $m_D$ : 1.875612945000000e+03
Relative diff: 1.23e-10 (9.9 digits, need 8)

[INFO] C5: H 1S-2S leading order (soft check)
Measured:      2466061413187018 Hz
(3/4)cR_inf: 2467381470187486 Hz
Rel diff:      5.35e-04 (3.3 digits)

Note: ~5 digit agreement expected (Lamb shift, QED, recoil)

GROUP D: KOIDE DERIVED ENTRIES (K1-K16)

```



```

-----
[PASS] D1: Koide  $K(e,\mu,\tau)$  from masses
From  $m_e, m_\mu, m_\tau$ : 6.666605114655210e-01
Stored K1: 6.666605115000001e-01
Relative diff: 5.17e-11 (10.3 digits, need 6)
[PASS] D2:  $a(\text{leptons})$  from K
Computed: 1.414200505100000e+00
Stored K4: 1.414200505200000e+00
Relative diff: 7.07e-11 (10.2 digits, need 6)
[PASS] D3:  $M(\text{leptons}) = (\text{sum sqrt } m)/3$ 
Computed: 1.771556100000000e+01
Stored K14: 1.771600000000000e+01
Relative diff: 2.48e-05 (4.6 digits, need 4)
[PASS] D4: Koide  $K(u,c,t)$  from masses
Computed: 8.487935475760000e-01
Stored K2: 8.487935476000000e-01
Relative diff: 2.83e-11 (10.5 digits, need 3)
[PASS] D5: Koide  $K(d,s,b)$  from masses
Computed: 7.312875768270000e-01
Stored K3: 7.312875768000000e-01
Relative diff: 3.69e-11 (10.4 digits, need 3)
GROUP E: EXACT SI CONSTANTS
-----

```

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[PASS] E1:  $h \cdot dv_{Cs}$  product consistency  
Computed: 6.091102297113866e-24  
Stored: 6.091102297113866e-24  
Agreement: EXACT

[PASS] E2: c is exact integer 299792458  
Computed: 2.9979245800000000e+08  
Stored: 2.9979245800000000e+08  
Agreement: EXACT

[PASS] E3: h exact SI 2019 value  
Computed: 6.6260701500000000e-34  
Stored: 6.6260701500000000e-34  
Agreement: EXACT

[PASS] E4: e exact SI 2019 value  
Computed: 1.6021766340000000e-19  
Stored: 1.6021766340000000e-19  
Agreement: EXACT

[PASS] E5:  $k_B$  exact SI 2019 value  
Computed: 1.3806490000000000e-23  
Stored: 1.3806490000000000e-23  
Agreement: EXACT

[PASS] E6:  $N_A$  exact SI 2019 value  
Computed: 6.0221407600000000e+23

Stored: 6.022140760000000e+23

Agreement: EXACT

GROUP F: LATTICE RATIO INDEPENDENCE — FINDING 15

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-----

Entries D9 ( $m_c/m_s = 11.783$ ), D10 ( $m_b/m_c = 4.578$ ),  
D11 ( $m_u/m_d = 0.485$ ) are INDEPENDENT measurements from  
lattice QCD (FLAG averages) evaluated at a COMMON scale.  
They are NOT derivable from individual PDG quark masses  
(D1-D5) which are evaluated at DIFFERENT scales:

$m_u, m_d, m_s$  at  $\mu = 2 \text{ GeV MS-bar}$

$m_c$  at  $\mu = m_c \sim 1.27 \text{ GeV}$

$m_b$  at  $\mu = m_b \sim 4.18 \text{ GeV}$

Discrepancies if naively compared:

$m_c/m_s$ : PDG gives 13.615, lattice = 11.783 (15.5%)

$m_b/m_c$ : PDG gives 3.286, lattice = 4.578 (28.2%)

$m_u/m_d$ : PDG gives 0.460, lattice = 0.485 (5.2%)

This is a renormalization scale mismatch, NOT a database error.

Finding 15 registered: [HOWL-DATA-4-FINDING-15]

GROUP G: GUT AND UNIFICATION VERIFICATION

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[PASS] G1: SM gap ratio = 218/115

Computed: 1.895652173913043e+00

Stored: 1.895652173913043e+00

Agreement: EXACT

[PASS] G2: SM+VL gap ratio = 38/27

Computed: 1.407407407407407e+00

Stored: 1.407407407407407e+00

Agreement: EXACT

[PASS] G3: MSSM gap ratio = 7/5

Computed: 1.400000000000000e+00

Stored: 1.400000000000000e+00

Agreement: EXACT

[PASS] G4: Modified b1 = 25/6

Computed: 4.166666666666667e+00

Stored: 4.166666666666667e+00

Agreement: EXACT

[PASS] G5: Modified b2 = -13/6

Computed: -2.166666666666667e+00

Stored: -2.166666666666667e+00

Agreement: EXACT

[PASS] G6: Modified b3 = -20/3

Computed: -6.666666666666667e+00

Stored: -6.666666666666667e+00

Agreement: EXACT

GROUP H: CABIBBO DOUBLET STAGED ENTRIES

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Entry 124: M_VL (Cabibbo Doublet mass)
Status: STAGED
Window: 1500000 - 6000000 MeV (1.5 - 6.0 TeV)
Sources: PHYS-16, PHYS-19
Bounds: LHC pair production (lower), CKM perturbativity (upper)

Entry 125: sin(theta_14) ~ |V_ub'|
Status: STAGED
Estimate: ~0.045 (from CKM first-row deficit)
Source: PHYS-19 (Belfatto, Berezhiani 2020)

Entry 126: theta_24 (2nd-gen mixing)
Status: STAGED (CONSTRAINED_BY_KAON_PHYSICS)

Entry 127: sin(theta_34) (3rd-gen mixing)
Status: STAGED (FROM_A_FB_B_FIT)

Entry 128: delta_1 (new CP phase)
Status: STAGED (CONSTRAINED_BY_NEDM)

Entry 129: delta_2 (new CP phase)
Status: STAGED (CONSTRAINED_BY_B_PHYSICS)

GROUP T: COMPUTATION TRACEABILITY MAP

```

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Paper    DATA-4 Entries Used          Computation
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PHYS-9 B1(alpha\_inv), G1(pi), G8(zeta3) alpha -  
 > a\_e via QED series  
  
 PHYS-12 B1,B11,C6,C4,C5,C1,B12 7 EW inputs -  
 > 11 observables  
  
 PHYS-13 B1,B11,B12 3 couplings -  
 > gap ratio 218/115 vs 1.358  
  
 PHYS-15 B1,B11,B12 Elimination cascade -  
 > (3,2,1/6)  
  
 PHYS-17 N1-N3 (SM betas) Generation democracy, boson prob  
 PHYS-18 N4-N6 (VL shifts) Y=1/6 asymmetry, 1/Y^2 scaling  
 PHYS-19 D6-D8 (CKM), B11 Three anomalies -  
 > (3,2,1/6)  
  
 PHYS-20 N7-N11 (modified betas, gap ratio) M\_GUT -  
 > proton decay tau ~ 10^34-35  
  
 PHYS-22 G10(pi^2), G8(zeta3), G3(ln2) A2 = 197/144 + (3/4)z3 + R4\*c  
 PHYS-23 B2-B4(lepton masses), D1-D5(quarks) Koide K all three sectors  
 PHYS-24 B1,B11,B12, N1-N6, N14(b\_ij) Two-  
 loop unification, Delta = -0.40  
  
 MATH-6 G1-G14 (Q335 basis) PSLQ 82/82 null, Bessel zeros

## RESULTS

| ID | Test                          | Digits | Need | Result |
|----|-------------------------------|--------|------|--------|
| A1 | m_p/m_e direct vs m_p / m_e   | 12.2   | 11   | PASS   |
| A2 | m_mu/m_e direct vs m_mu / m_e | 21.0   | 10   | PASS   |

|    |                                                        |       |     |      |
|----|--------------------------------------------------------|-------|-----|------|
| A3 | $m_{\tau}/m_e$ direct vs $m_{\tau} / m_e$              | 6.3   | 6   | PASS |
| A4 | $m_{\tau}/m_{\mu}$ direct vs $m_{\tau} / m_{\mu}$      | 5.8   | 5   | PASS |
| A5 | $m_n/m_p$ direct vs $m_n / m_p$                        | 20.0  | 11  | PASS |
| A6 | $M_W/M_Z$ direct vs $M_W / M_Z$                        | 7.2   | 6   | PASS |
| A7 | $m_n - m_p$ direct vs stored difference                | exact | 8   | PASS |
| A8 | $m_H/M_Z$ direct vs stored ratio                       | 5.6   | 5   | PASS |
| A9 | $m_t/M_Z$ direct vs stored ratio                       | 5.9   | 5   | PASS |
| B1 | $R2 = \pi/4$ (Q335 vs mpmath@120)                      | 102.0 | 100 | PASS |
| B2 | $R4 = \pi^2/32$ (Q335 vs mpmath@120)                   | 102.4 | 100 | PASS |
| B3 | $2\pi = 8*R2$ (exact within Q335)                      | exact | 100 | PASS |
| B4 | $\zeta(2) = \pi^2/6$ (Q335 numerators)                 | 101.5 | 99  | PASS |
| B5 | $\alpha/\pi$ stored vs $1/(\alpha_{\text{inv}} * \pi)$ | 13.9  | 12  | PASS |
| B6 | $\phi = (1 + \sqrt{5})/2$ (Q335)                       | 101.4 | 100 | PASS |
| B7 | $\pi^2$ stored vs $\pi * \pi$ (Q335)                   | 101.7 | 99  | PASS |
| B8 | $2\pi$ stored vs $2 * \pi$ (exact)                     | exact | 100 | PASS |
| C1 | $R_{\text{inf}} = \alpha^2 m_e c / (2h)$               | 11.3  | 11  | PASS |
| C2 | $a_0 = \hbar / (m_e c \alpha)$                         | 11.4  | 11  | PASS |
| C3 | $\mu_0 = 2*\alpha*h / (c*e^2)$                         | 11.9  | 11  | PASS |
| C4 | $m_D = m_p + m_n - E_D$                                | 9.9   | 8   | PASS |
| D1 | Koide $K(e, \mu, \tau)$ from masses                    | 10.3  | 6   | PASS |
| D2 | $a(\text{leptons})$ from $K$                           | 10.2  | 6   | PASS |
| D3 | $M(\text{leptons}) = (\sum \sqrt{m})/3$                | 4.6   | 4   | PASS |
| D4 | Koide $K(u, c, t)$ from masses                         | 10.5  | 3   | PASS |

|    |                                   |       |    |      |
|----|-----------------------------------|-------|----|------|
| D5 | Koide $K(d,s,b)$ from masses      | 10.4  | 3  | PASS |
| E1 | $h * dv_{Cs}$ product consistency | exact | 15 | PASS |
| E2 | $c$ is exact integer 299792458    | exact | 15 | PASS |
| E3 | $h$ exact SI 2019 value           | exact | 9  | PASS |
| E4 | $e$ exact SI 2019 value           | exact | 10 | PASS |
| E5 | $k_B$ exact SI 2019 value         | exact | 7  | PASS |
| E6 | $N_A$ exact SI 2019 value         | exact | 9  | PASS |
| G1 | SM gap ratio = 218/115            | exact | 10 | PASS |
| G2 | SM+VL gap ratio = 38/27           | exact | 10 | PASS |
| G3 | MSSM gap ratio = 7/5              | exact | 10 | PASS |
| G4 | Modified $b_1 = 25/6$             | exact | 10 | PASS |
| G5 | Modified $b_2 = -13/6$            | exact | 10 | PASS |
| G6 | Modified $b_3 = -20/3$            | exact | 10 | PASS |

Total: 38 tests, 38 PASS, 0 FAIL

Inherited from DATA-3: 32 checks (Groups A-E)

New in DATA-4: 6 checks (Group G: GUT verification)

ENTRY COUNT:

Sections A-K (inherited from DATA-3): 123 entries

Section L (Cabibbo Doublet, STAGED): 6 entries

Section N (GUT parameters, Level 1): 17 entries

Total: 146 entries

+=====+

| ALL TESTS PASS

|



|  |                                                        |  |
|--|--------------------------------------------------------|--|
|  |                                                        |  |
|  | DATA-4 DECLARATION:                                    |  |
|  | DATA-3 (123 entries, 32/32 checks) inherited.          |  |
|  | Finding 15 (lattice independence) formalized.          |  |
|  | 6 Cabibbo Doublet parameters STAGED (entries 124-129). |  |
|  | 17 GUT/unification parameters added (Section N).       |  |
|  | 6 new GUT verification checks added (Group G).         |  |
|  | Computation traceability map added (Group T).          |  |
|  |                                                        |  |
|  | DATA-3 is retired.                                     |  |
|  | All future HOWL computation references DATA-4.         |  |

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DATA-4 VERIFICATION COMPLETE

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[\[HOWL-DATA-4-2026\]](https://github.com/ghowland/papers/tree/main/papers/2026/04/01/verified-integer-rational-database) Verified Integer Rational Database . Github: <https://github.com/ghowland/papers/tree/main/papers/2026/04/01/verified-integer-rational-database>  
 DATA-4-2026

[\[HOWL-DATA-1-2026\]](https://github.com/ghowland/papers/tree/main/papers/2026/04/01/cross-domain-precision-data-inventory) Cross-Domain Precision Data Inventory. Github: <https://github.com/ghowland/papers/tree/main/papers/2026/04/01/cross-domain-precision-data-inventory>  
 DATA-1-2026

[\[HOWL-DATA-2-2026\]](https://github.com/ghowland/papers/tree/main/papers/2026/04/01/integer-rational-representations-in-q335) Integer Rational Representations in Q335. Github: <https://github.com/ghowland/papers/tree/main/papers/2026/04/01/integer-rational-representations-in-q335>  
 DATA/HOWL-DATA-2-2026

[\[HOWL-DATA-3-2026\]](https://github.com/ghowland/papers/tree/main/papers/2026/04/01/verified-integer-rational-database) Verified Integer Rational Database. Github: <https://github.com/ghowland/papers/tree/main/papers/2026/04/01/verified-integer-rational-database>  
 DATA-3-2026